

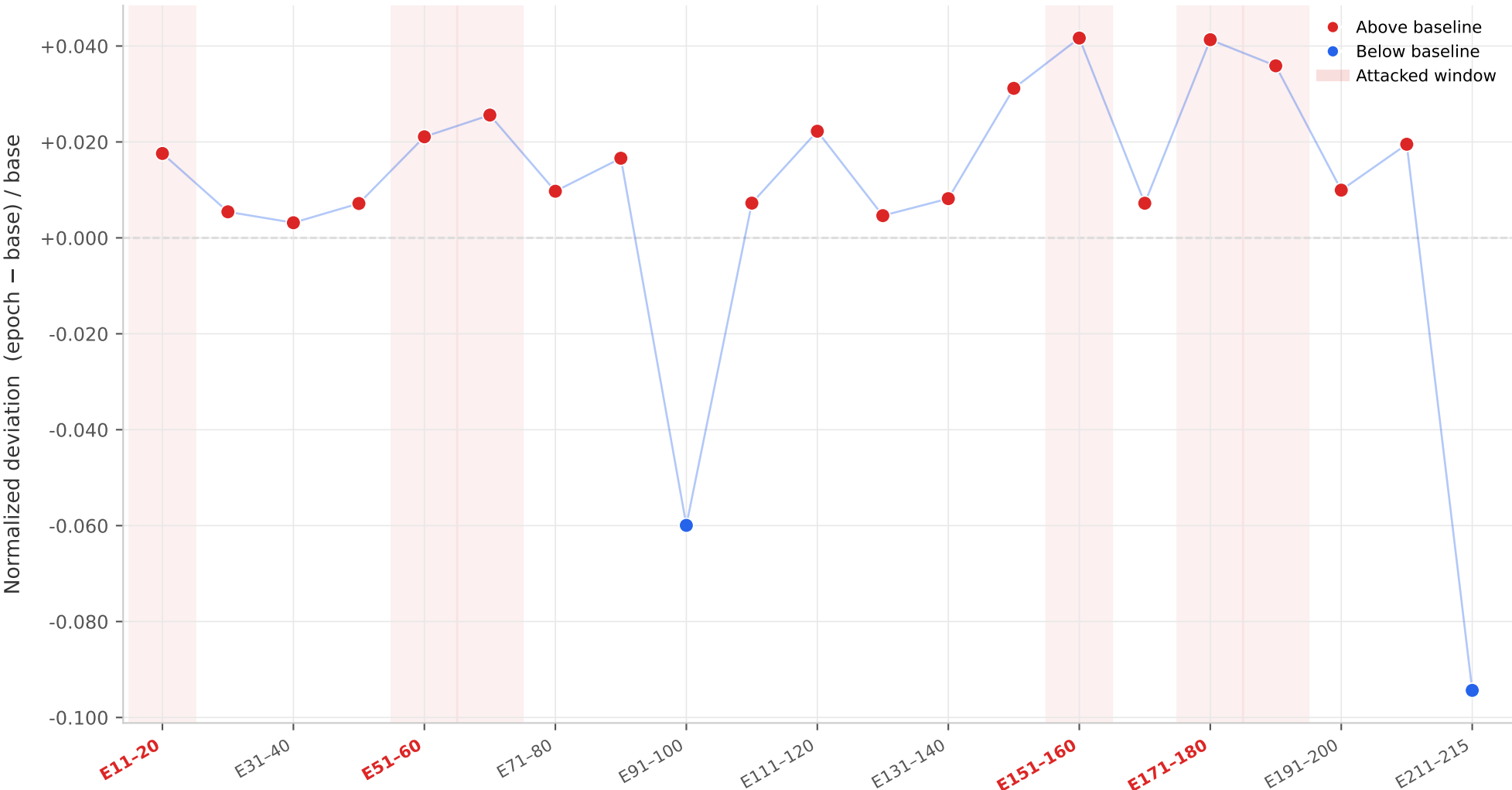
Entropy Stream Analysis

Deviation from Baseline — Per-Metric Report

Run Parameters		Baseline Values	
Dataset	epochs_5.0_1.0_sparse.json	Src IP Entropy	40.79
Directory	CAIDA_Dataset/txtD280/epochs/	Dst IP Entropy	32.08
Timestamp	2026-02-23T21:02:07.833224	Src Port Entropy	22.05
Total epochs	215	Dst Port Entropy	43.05
Baseline epochs	10 (5.0%)	Src+Dst IP Entropy	52.55
Attack epochs	205	Src+Dst Port Entropy	53.96
Malicious traffic	1.0%	Src IP Cardinality	5022
Epoch group size	10 epochs per point	Dst IP Cardinality	5005
Metrics plotted	20	Src Port Cardinality	4288
Mode	SPARSE	Dst Port Cardinality	5020
Attack probability	0.3	Src+Dst IP Cardinality	5040
Windows total	21	Src+Dst Port Cardinality	5037
Windows attacked	6	Src IP Uniformity	3.919e+08
Windows clean	15	Dst IP Uniformity	1.085e+06
		Src/Dst Ratio	1.004
		Total Flows	5056

Source IP Entropy

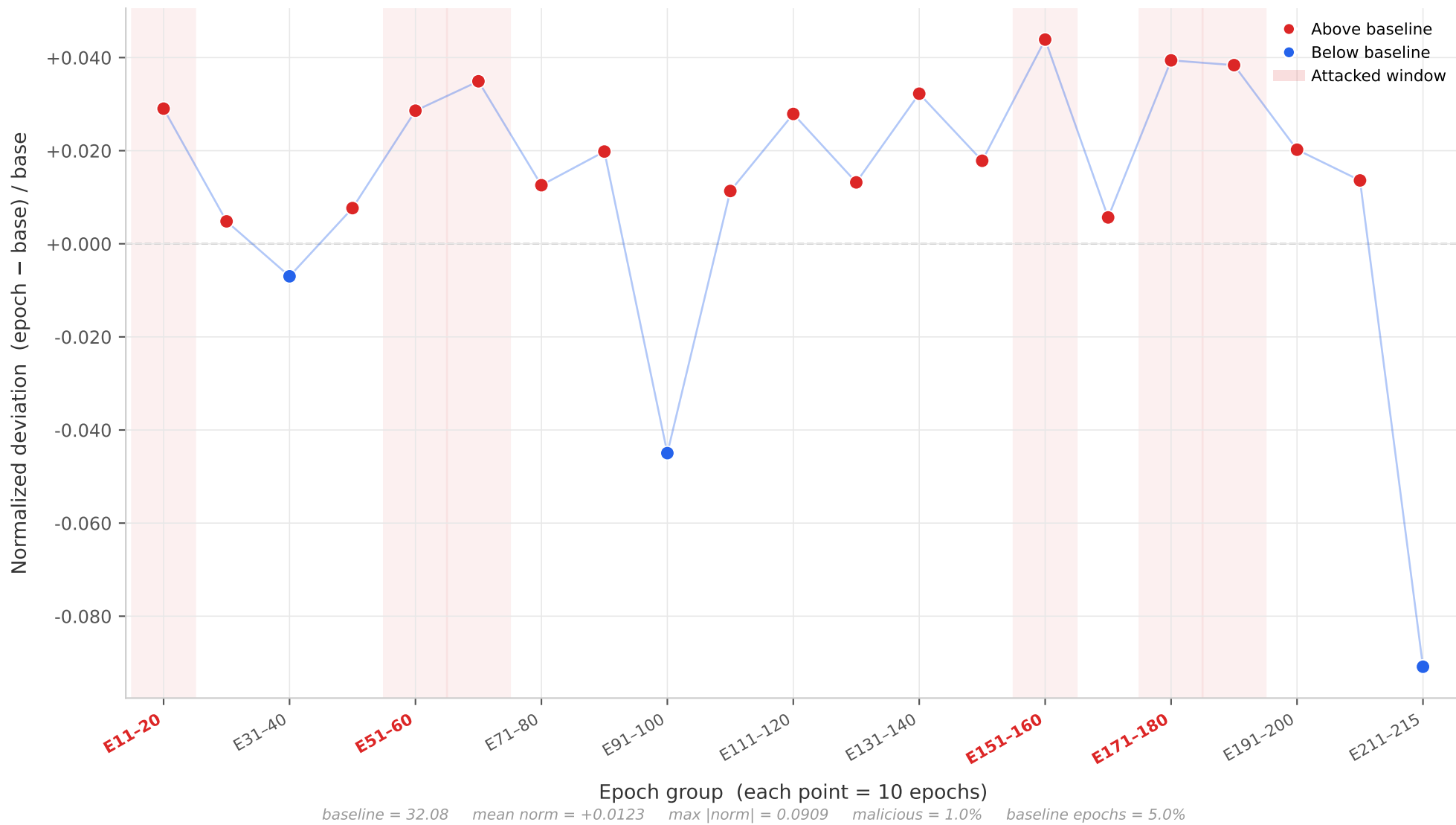
Entropy



baseline = 40.79 mean norm = +0.0086 max |norm| = 0.0943 malicious = 1.0% baseline epochs = 5.0%

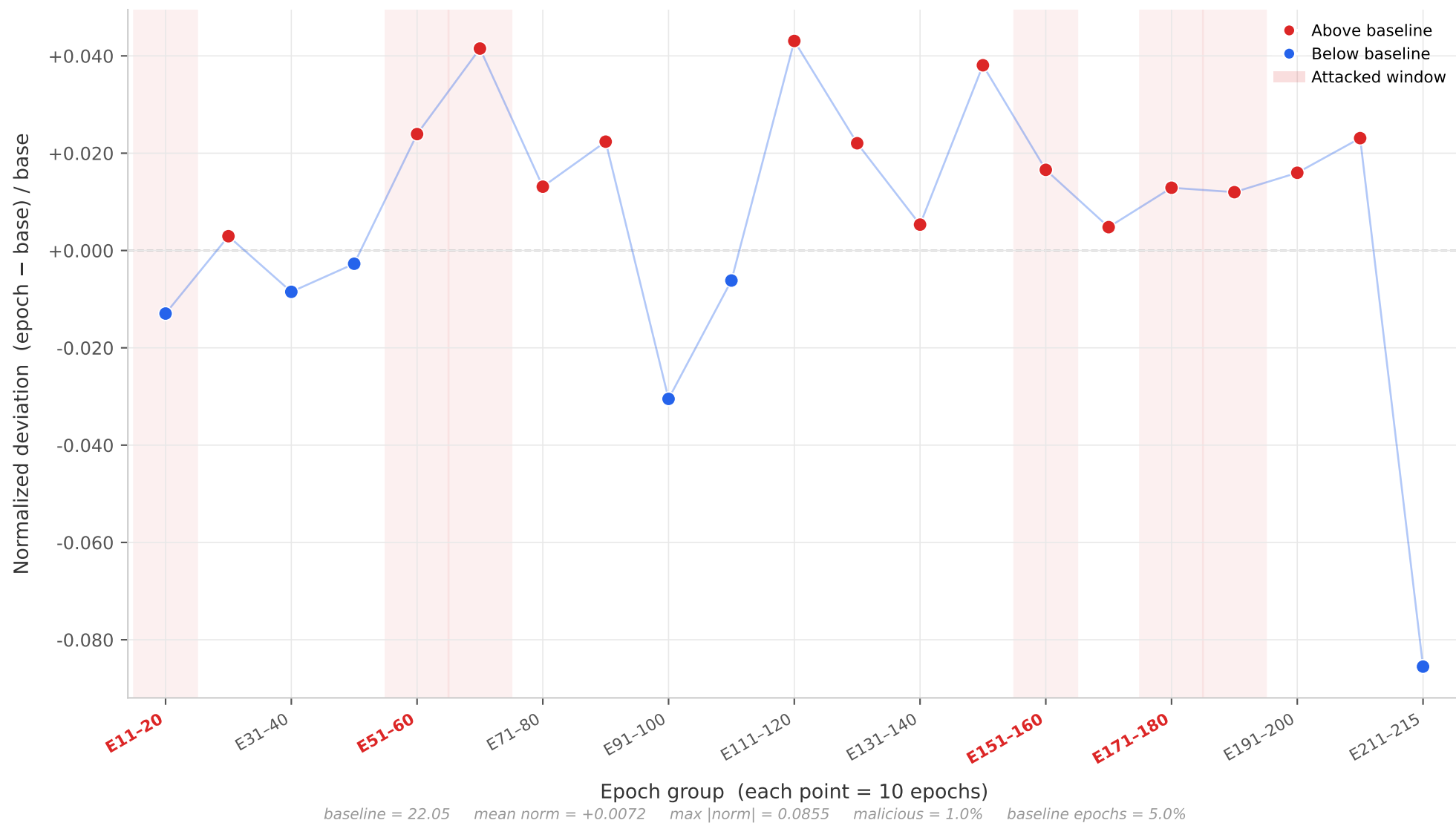
Destination IP Entropy

Entropy



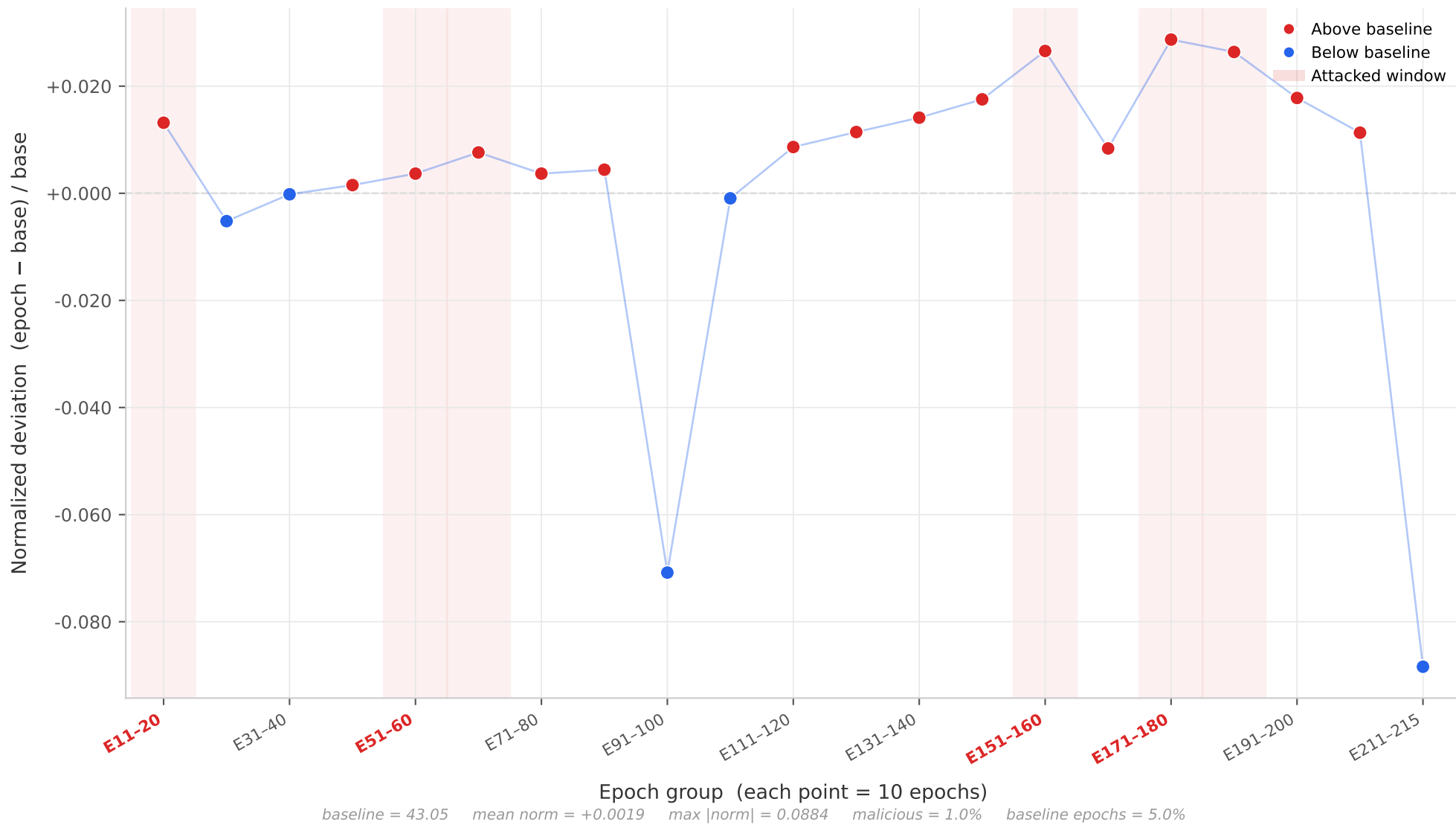
Source Port Entropy

Entropy



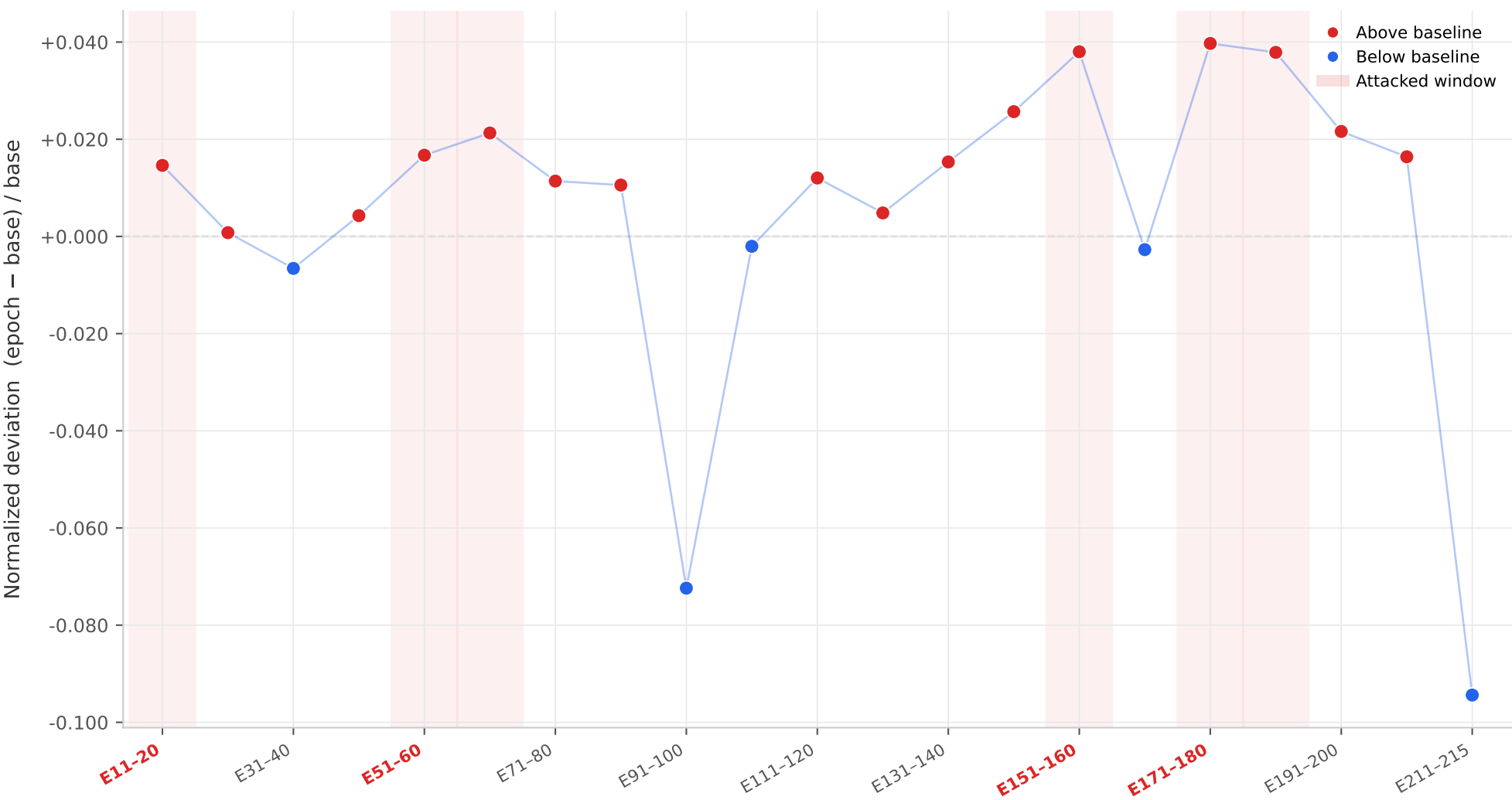
Destination Port Entropy

Entropy



Src+Dst IP Pair Entropy

Entropy

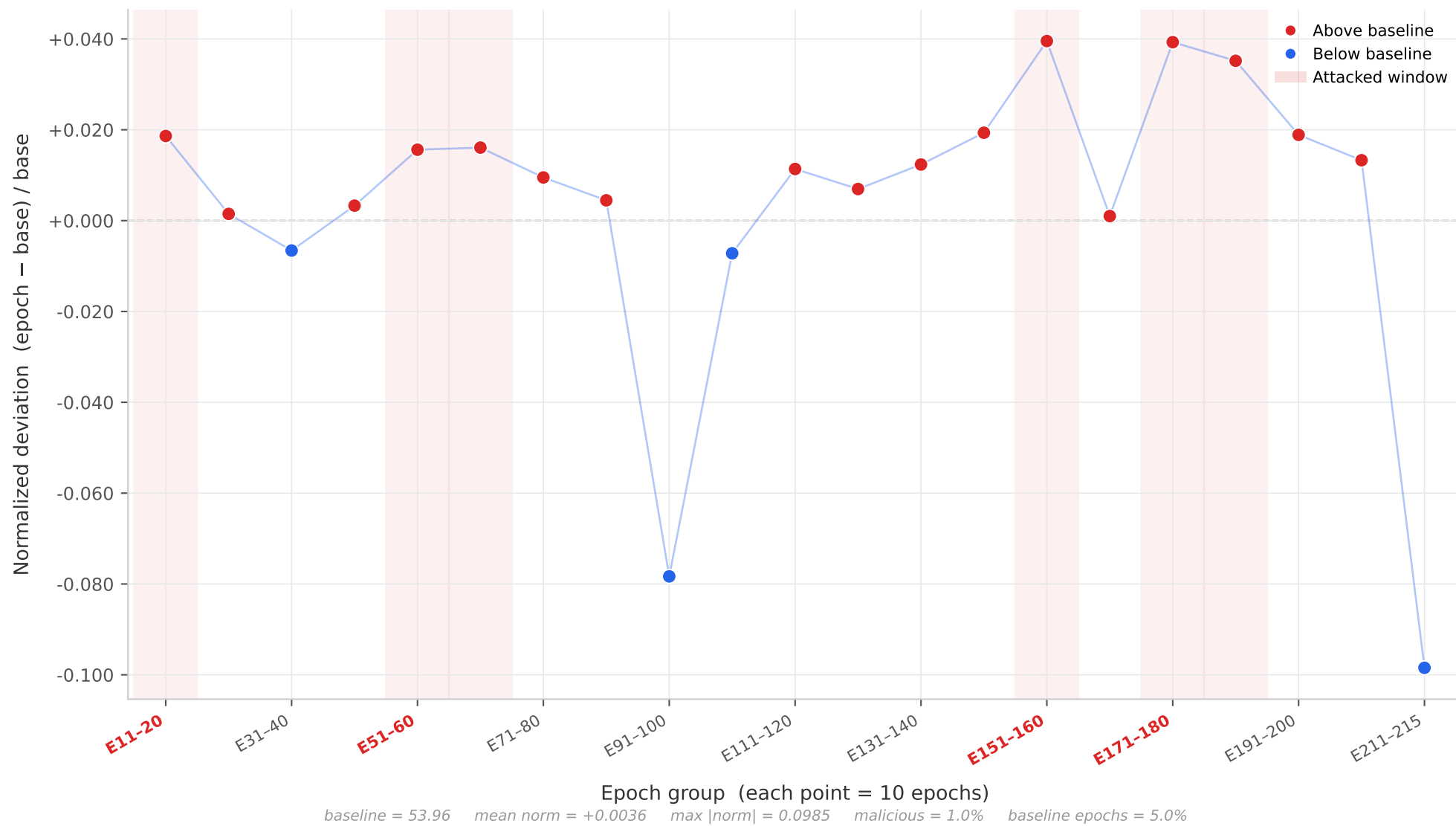


Epoch group (each point = 10 epochs)

baseline = 52.55 mean norm = +0.0054 max |norm| = 0.0944 malicious = 1.0% baseline epochs = 5.0%

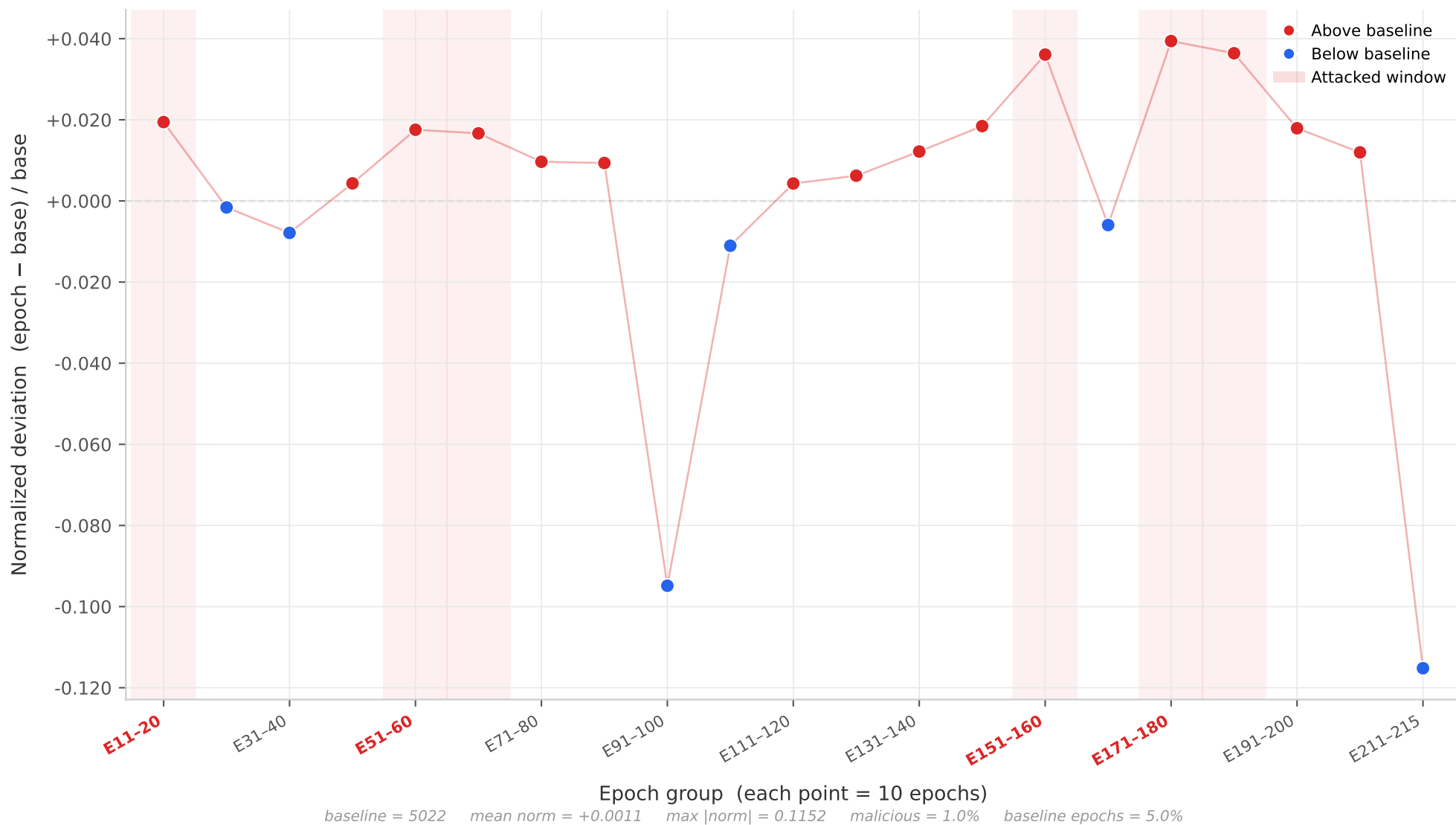
Src+Dst Port Pair Entropy

Entropy



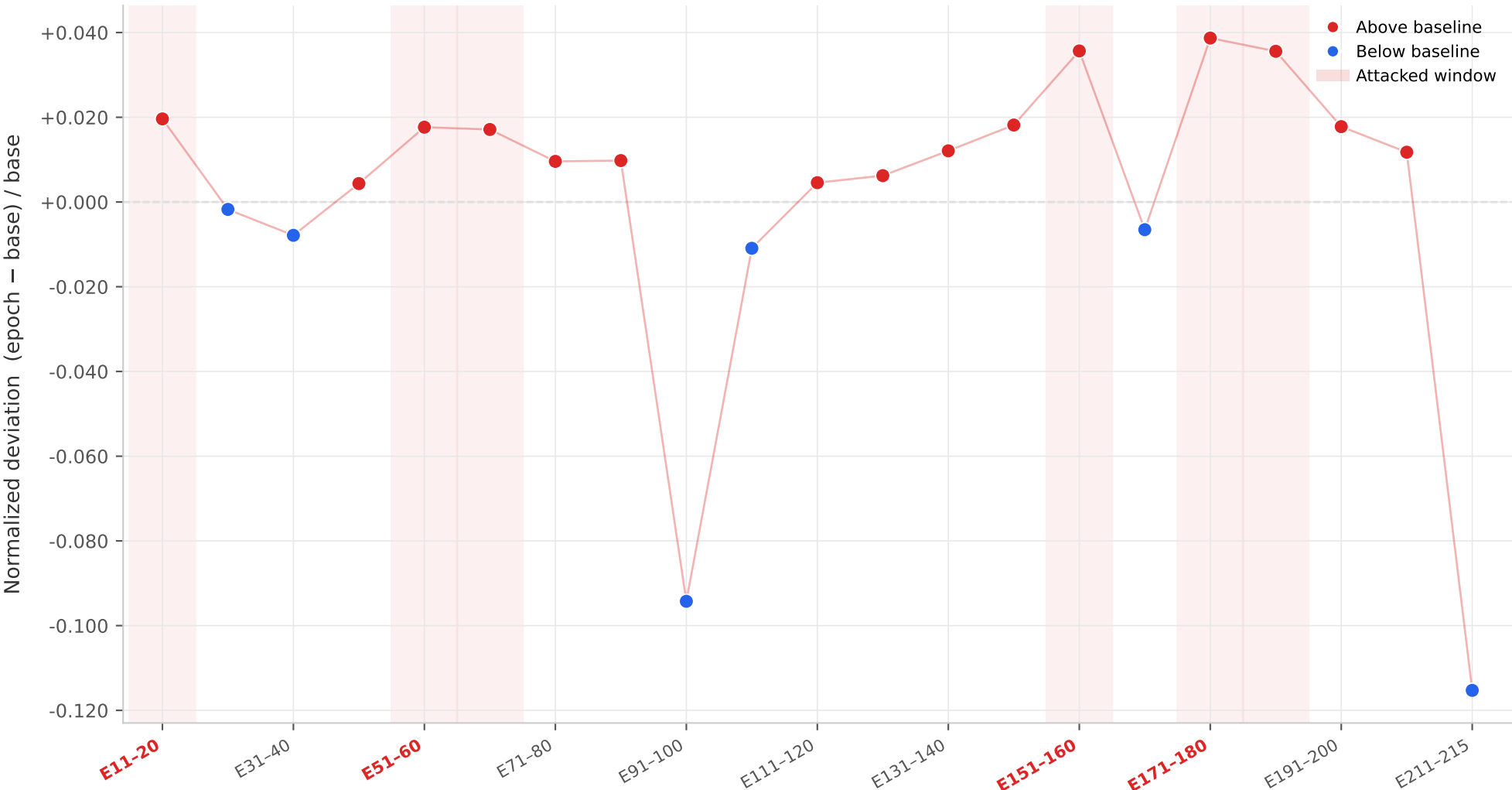
Source IP Cardinality

Cardinality



Destination IP Cardinality

Cardinality

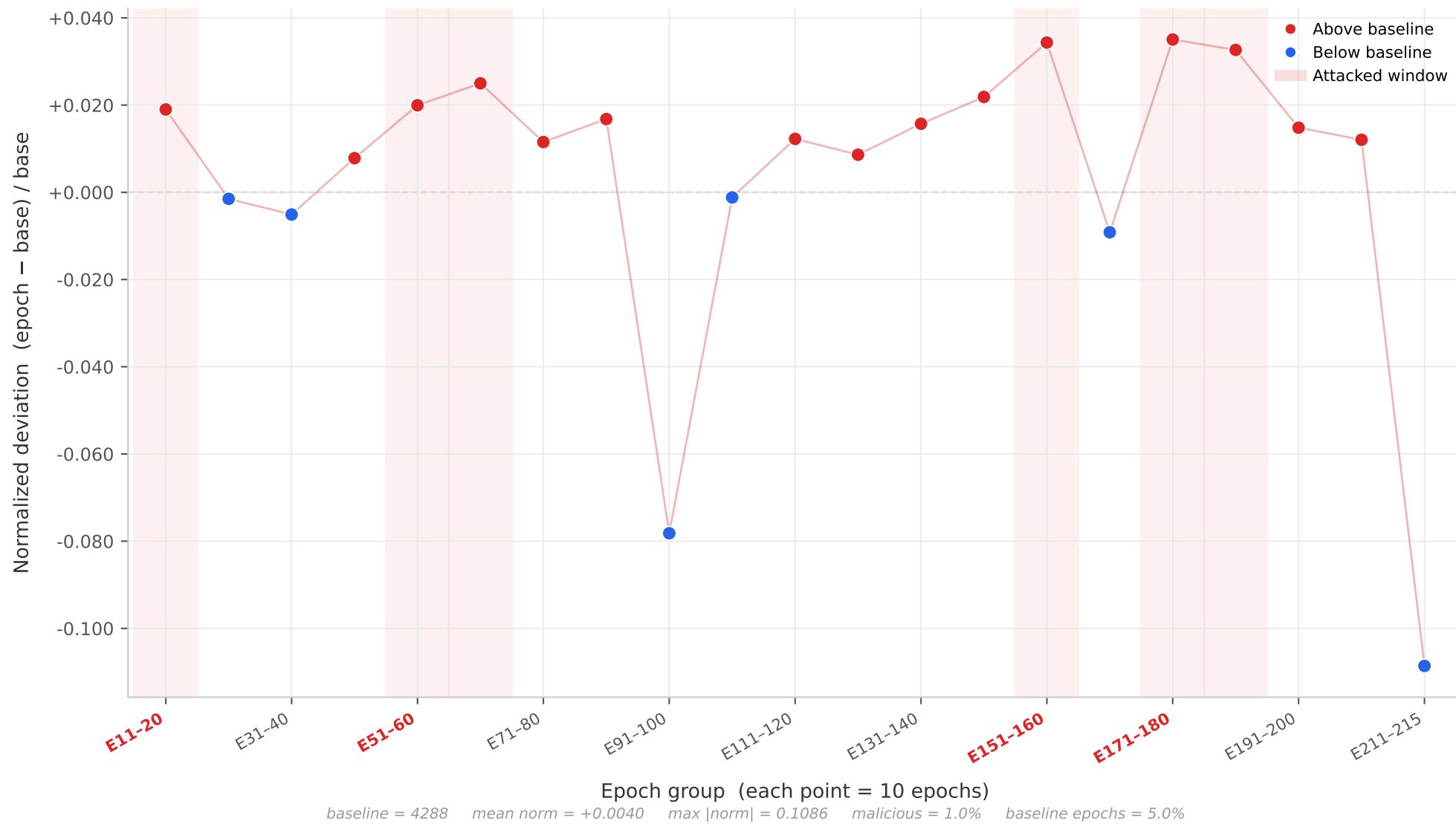


Epoch group (each point = 10 epochs)

baseline = 5005 mean norm = +0.0010 max |norm| = 0.1153 malicious = 1.0% baseline epochs = 5.0%

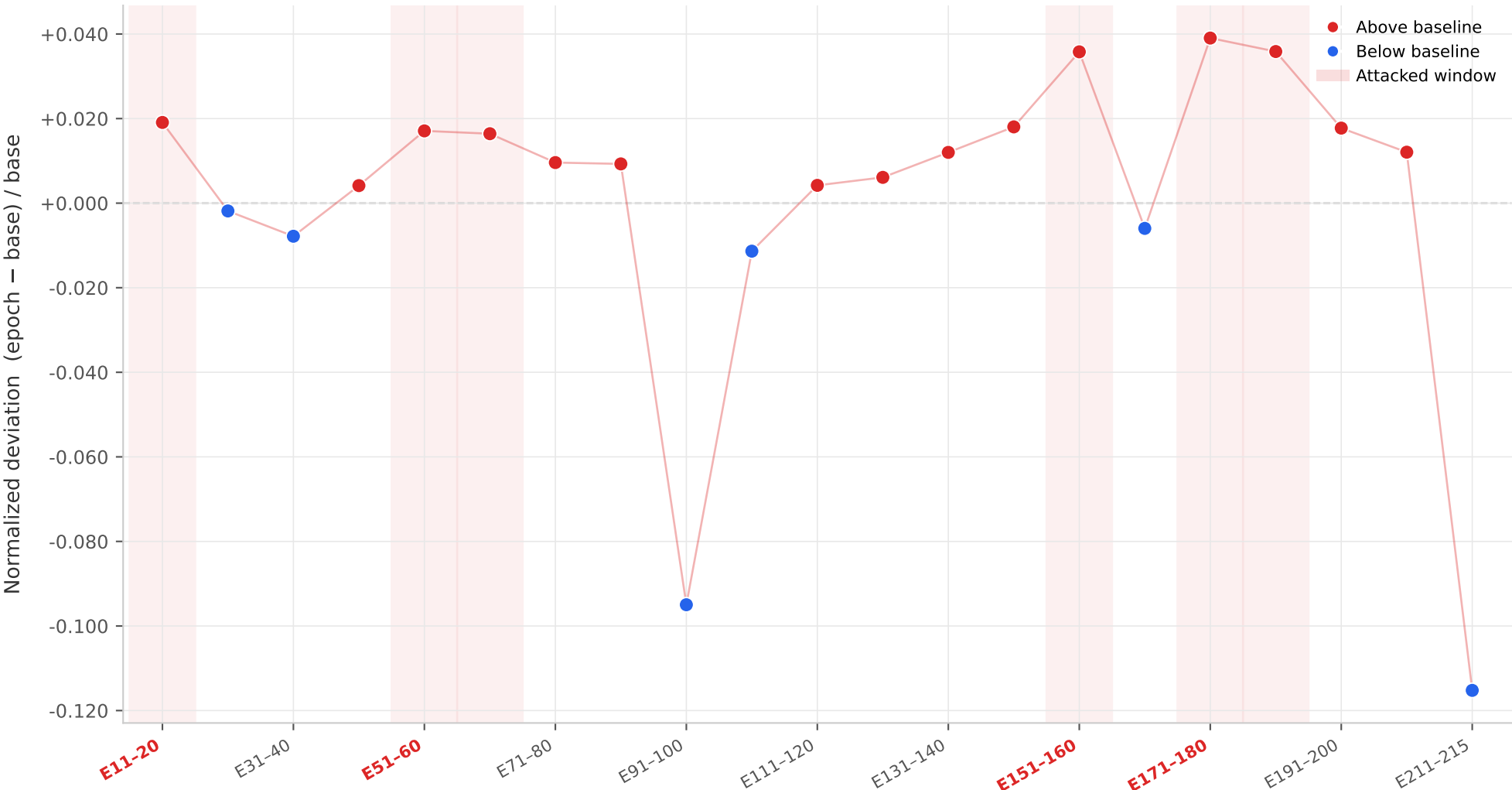
Source Port Cardinality

Cardinality



Destination Port Cardinality

Cardinality

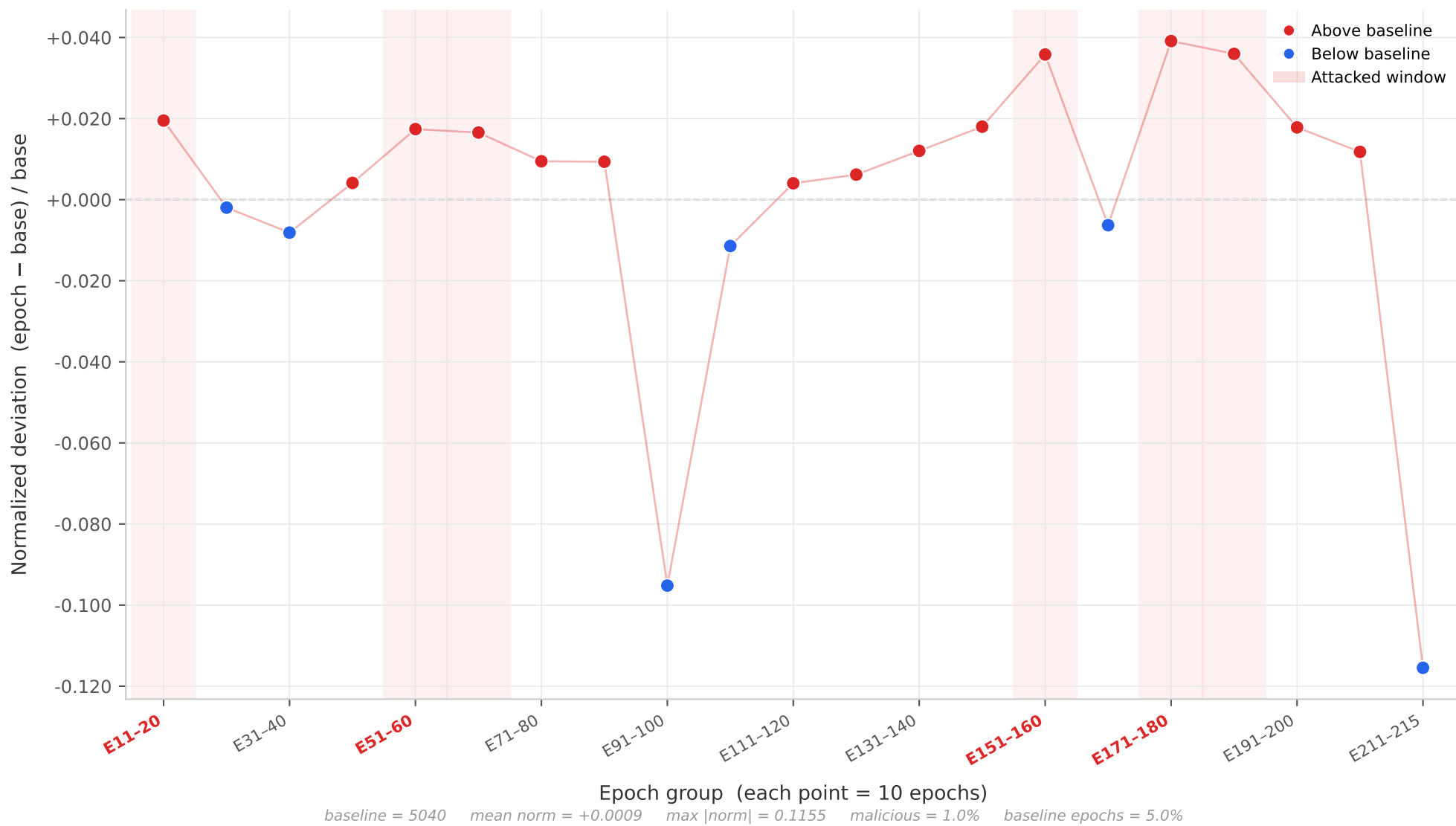


Epoch group (each point = 10 epochs)

baseline = 5020 mean norm = +0.0009 max |norm| = 0.1152 malicious = 1.0% baseline epochs = 5.0%

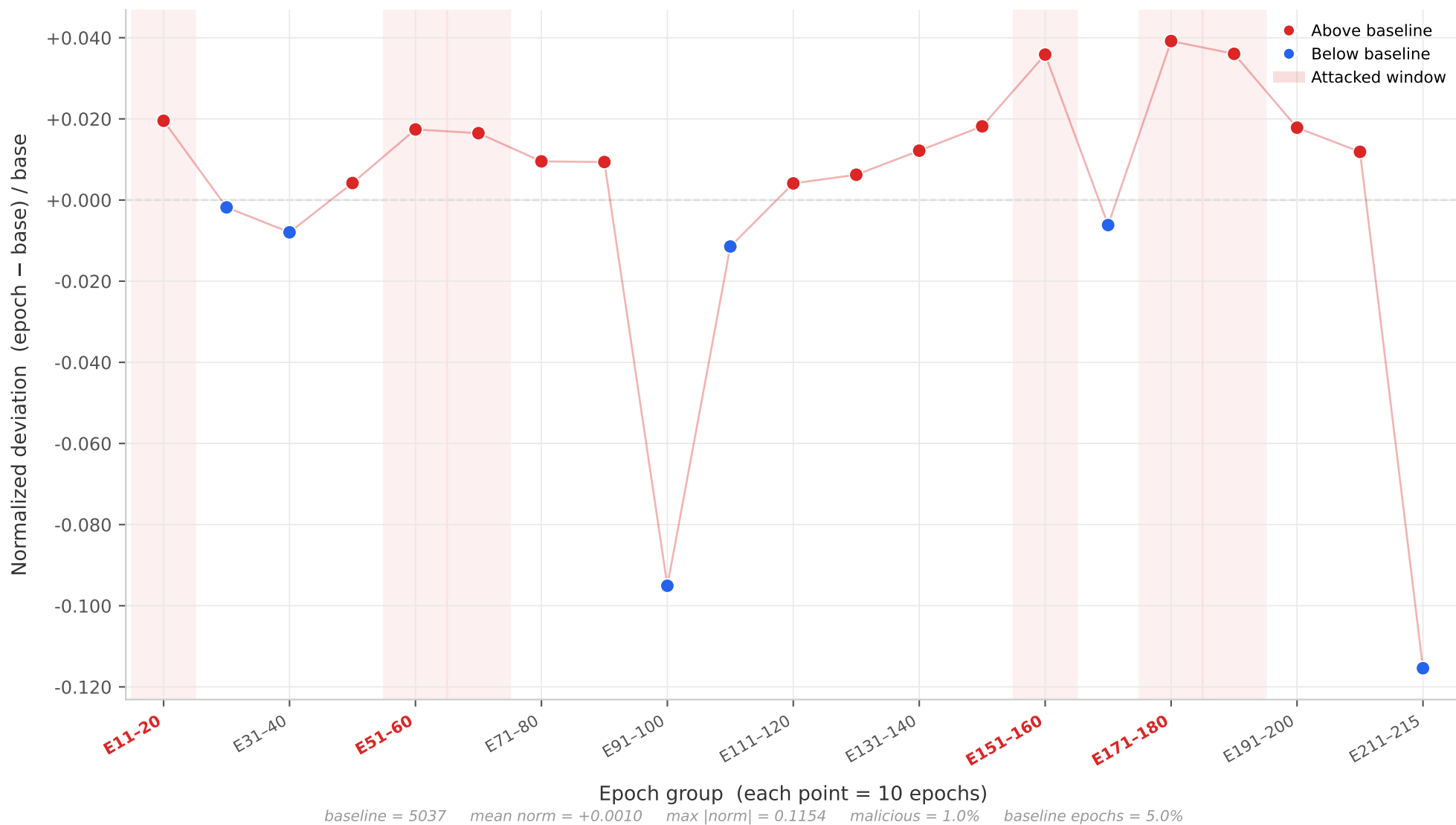
Src+Dst IP Pair Cardinality

Cardinality



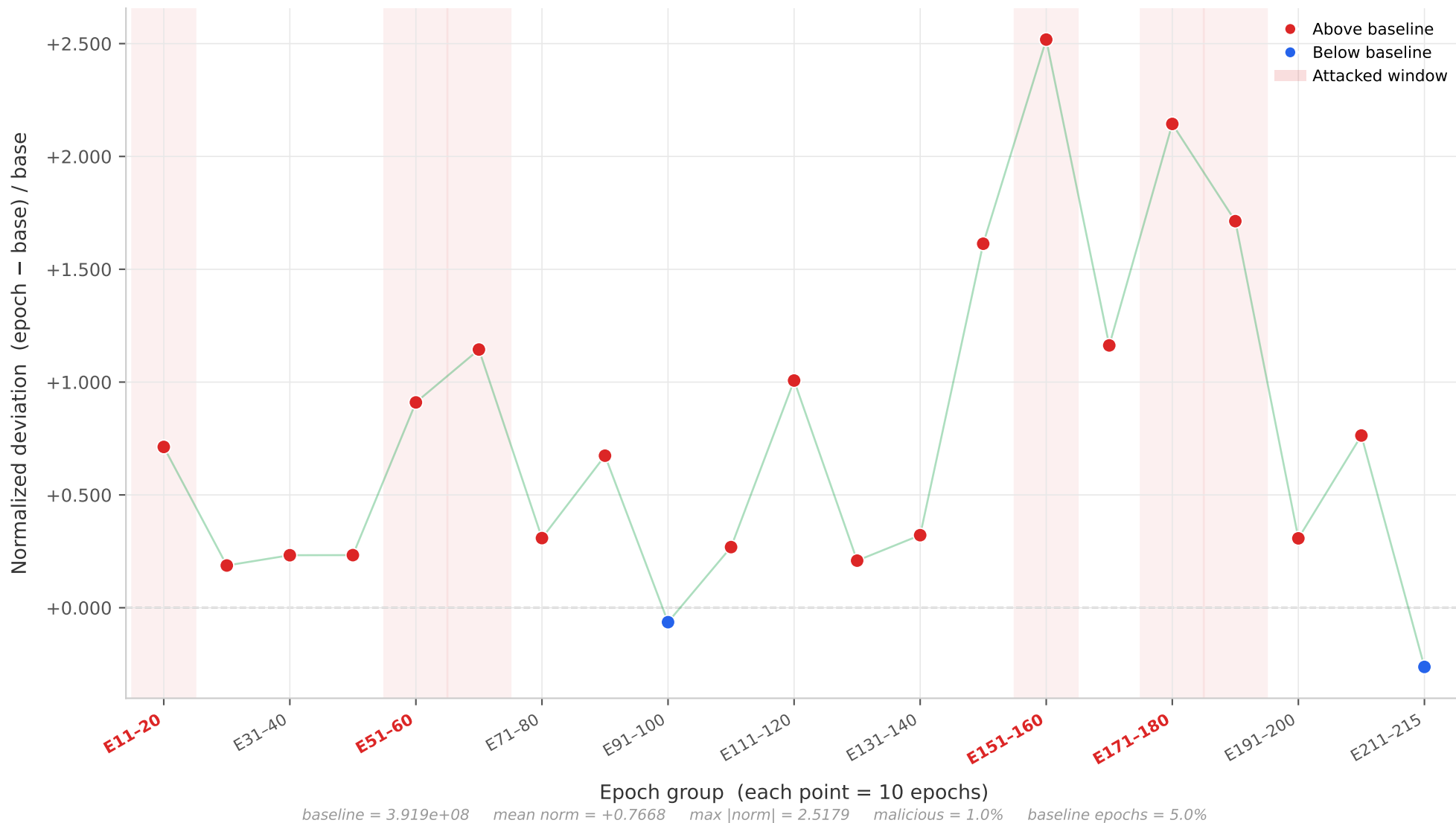
Src+Dst Port Pair Cardinality

Cardinality



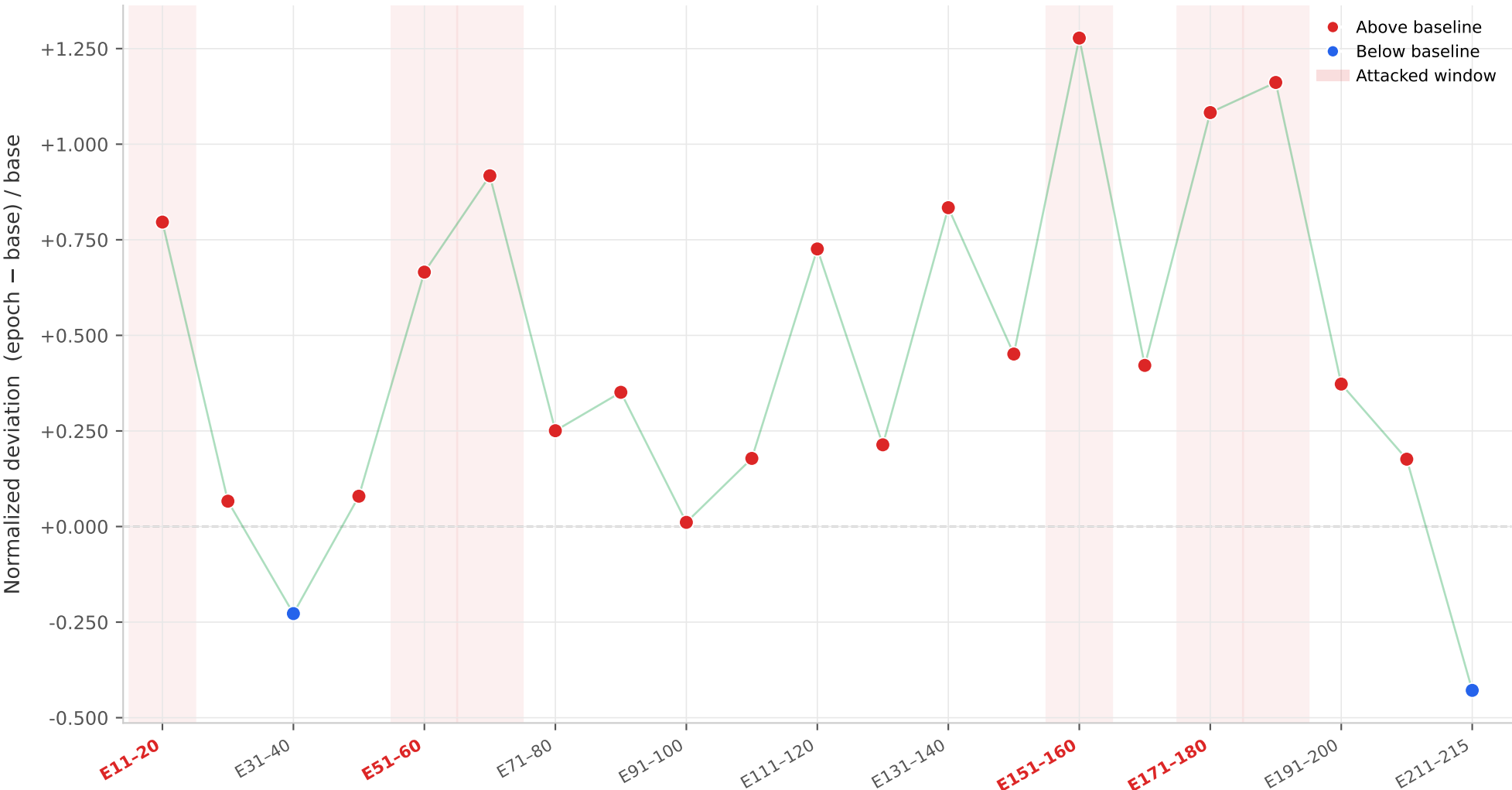
Source IP Uniformity

Uniformity



Destination IP Uniformity

Uniformity

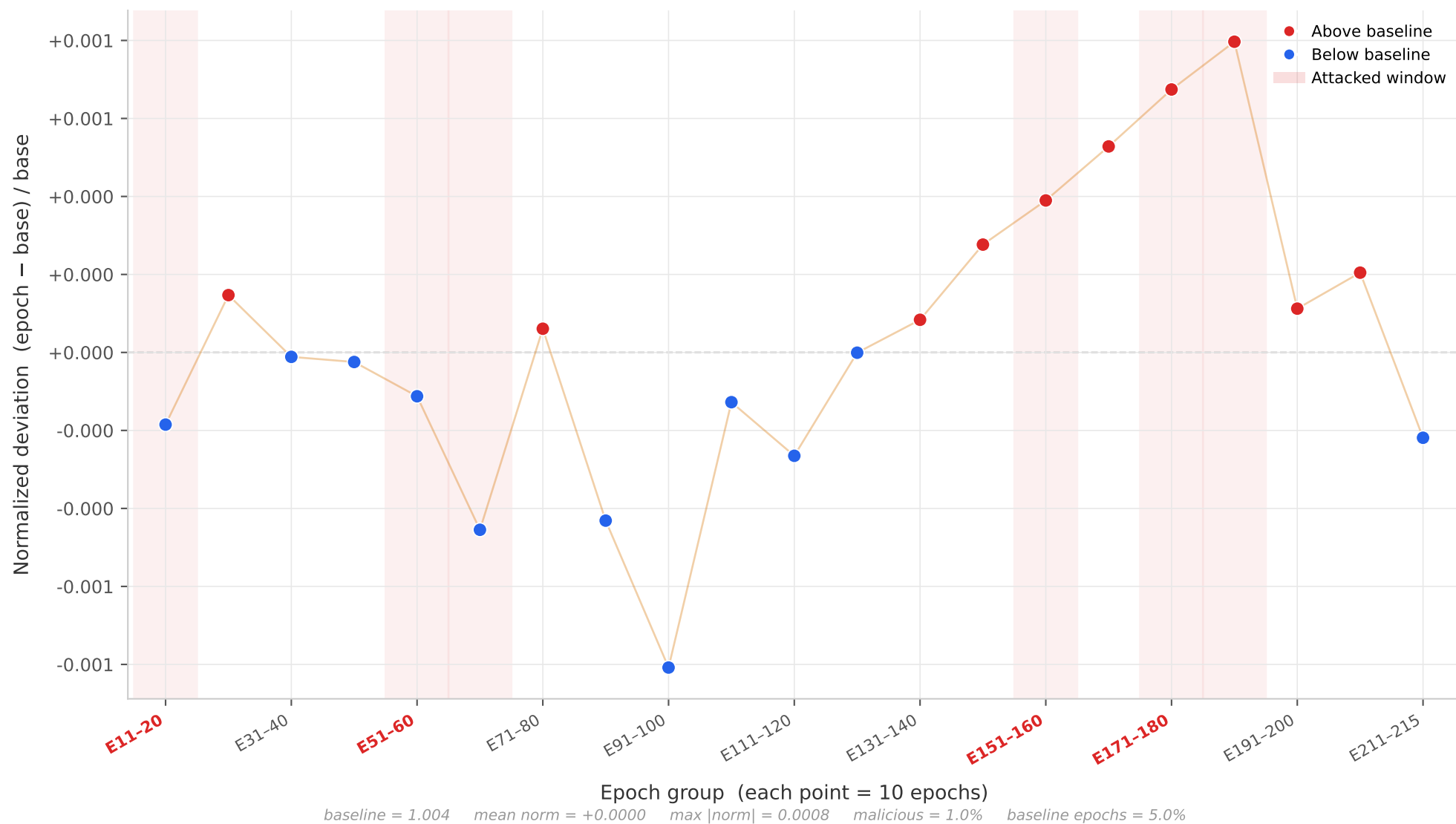


Epoch group (each point = 10 epochs)

baseline = 1.085e+06 mean norm = +0.4465 max |norm| = 1.2776 malicious = 1.0% baseline epochs = 5.0%

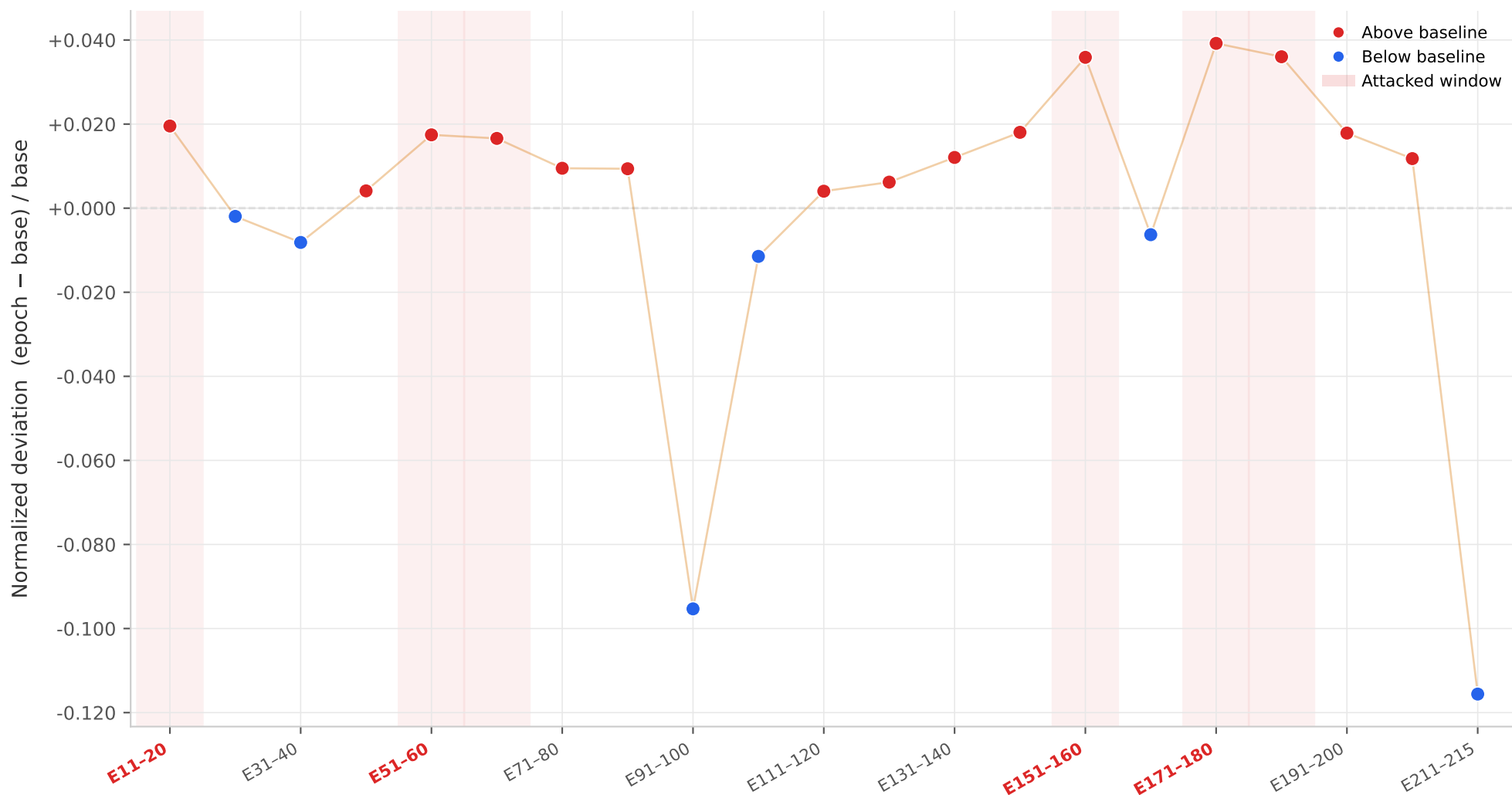
Src / Dst IP Ratio

Flow



Total Flows

Flow

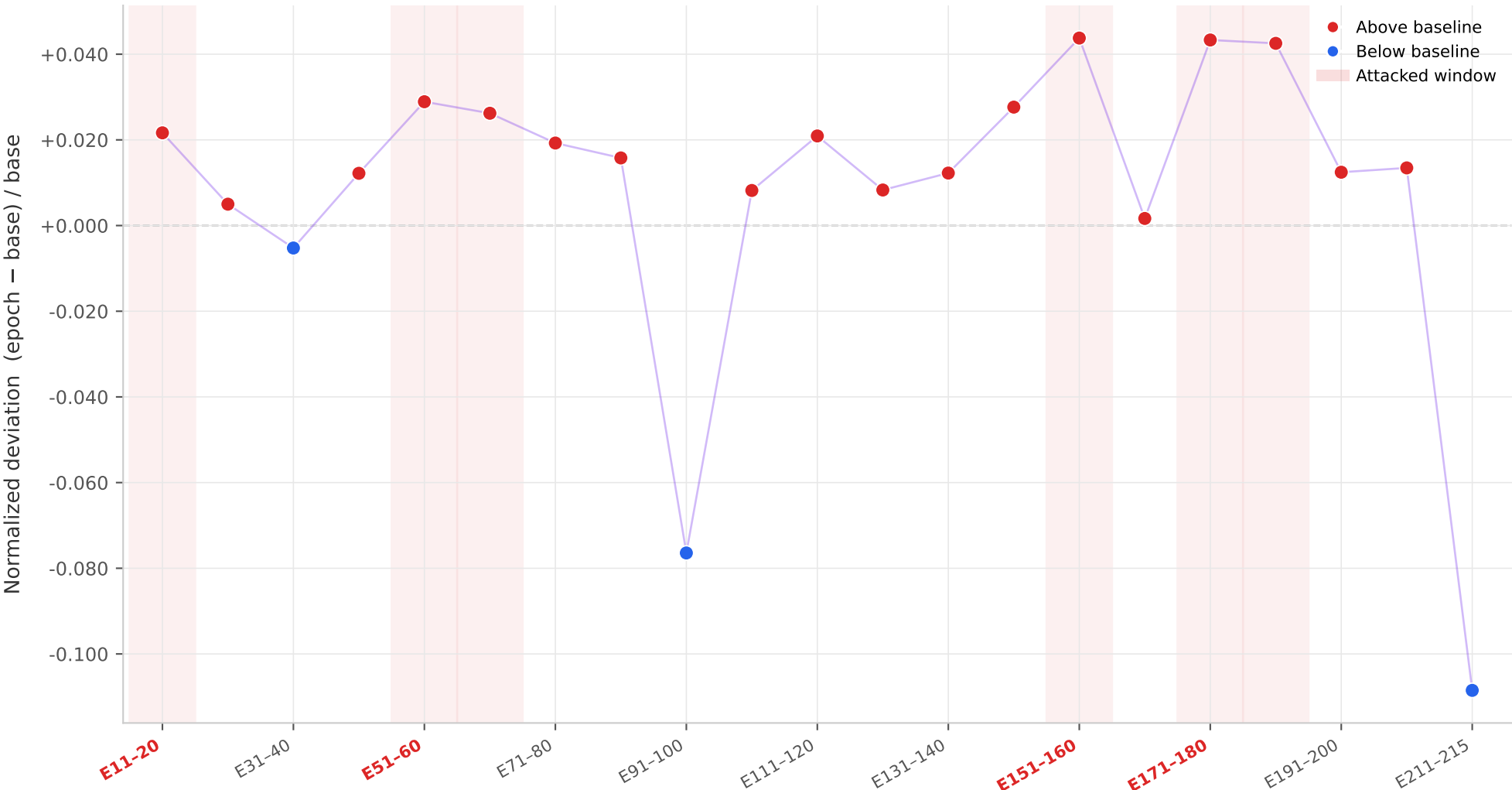


Epoch group (each point = 10 epochs)

baseline = 5056 mean norm = +0.0009 max |norm| = 0.1156 malicious = 1.0% baseline epochs = 5.0%

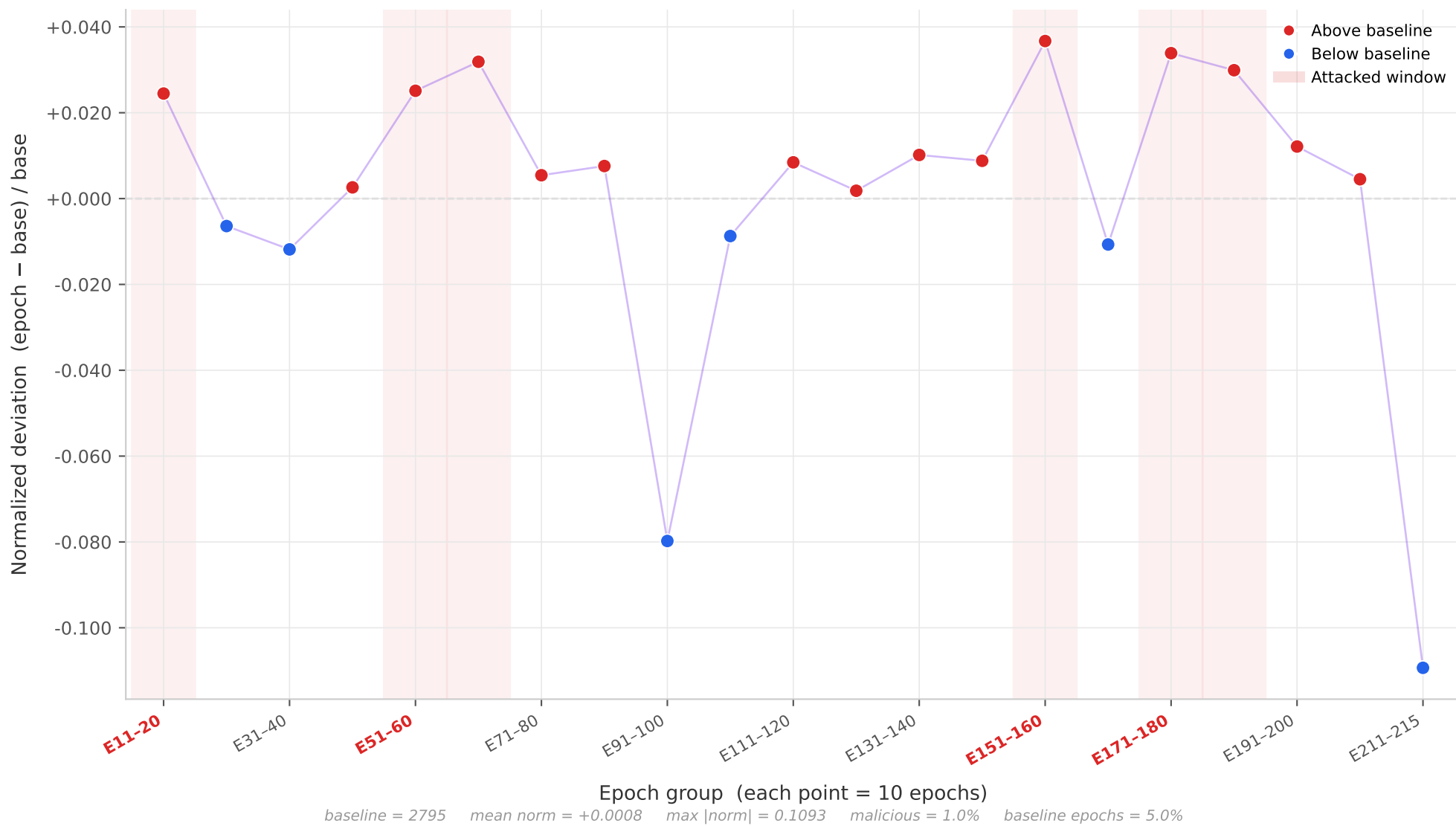
Unique Source IPs

Count



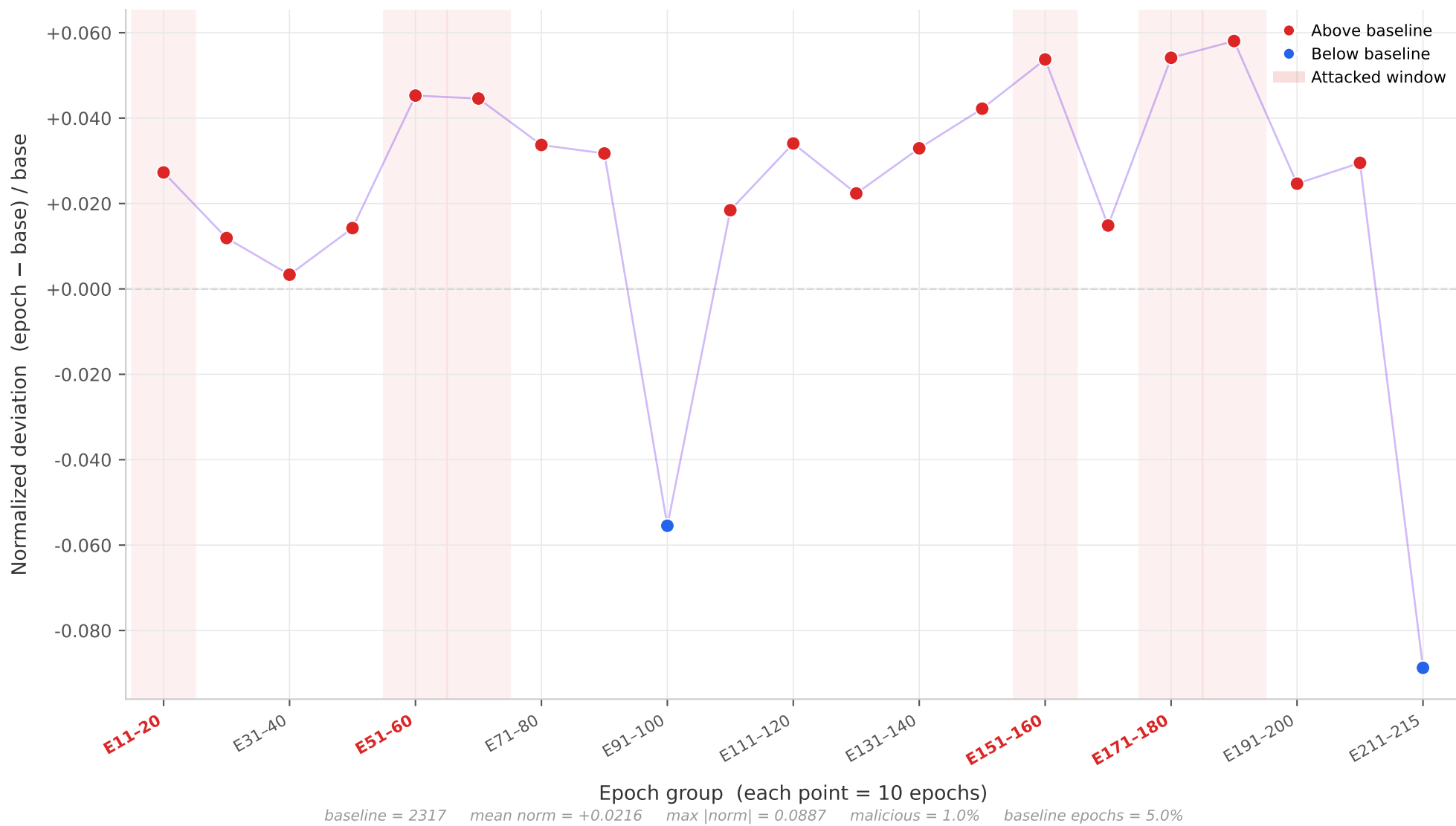
Unique Destination IPs

Count



Unique Source Ports

Count



Unique Destination Ports

Count

